Yaseen Khatib

Senior Full-Stack AI Engineer

I ship AI products end to end — six solo in twelve months, each with a link you can open.

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Hyderabad, India · Remote / Hybrid / On-site

6 — PRODUCTS SHIPPED SOLO — in twelve months

23M — PARAMETER MODEL DISTILLED — from a 9B teacher, runs on CPU

94% — PAYLOAD REDUCTION — React Flow serialization adapter

1.5–2x — FASTER DASHBOARDS — MongoDB pipeline profiling

93.5% — BACKEND TEST COVERAGE — FitRank API, CI on every release

THE STORY

I started in 2021 on **hospital information systems** — ASP.NET and React, real users every day — and learned how data moves through a system before I knew why that mattered. In 2023 I went **full-stack on MERN** at MSA Software, profiling slow MongoDB pipelines and moving the team to TypeScript and Next.js; ChatGPT had launched a few weeks earlier, so my nights became embeddings, retrieval and evals. By 2025 the two tracks merged: at Sparity I build the **LLM and agentic layer** of production systems by day, and on the side I have shipped **six products solo** — a distilled decision model, a Rust desktop app, a local-first finance agent, a workflow engine and two autonomous content pipelines — each with a link you can open.

SELECTED WORK

full write-ups on pages 2–3

FitRank

APPLIED ML · LIVE

23M MiniLM student distilled from a 9B teacher. Beats a rules-only baseline on hit@5 and pairwise ordering, served on CPU at $0 per decision.

fitrank.streamerosai.com ↗

streamerOS

DESKTOP · RUST

Rust + Tauri 2 desktop cockpit for streamers, written with Claude Code. 24k lines of Rust, 306 Rust tests, OBS automation, local LLM. Launching November 2026.

streamerosai.com ↗

EXPERIENCE

Senior Software Developer
06/2025 – Present

Sparity · Hyderabad · Hybrid

- Own the **LLM and agentic layer** of production MERN systems end to end — architecture, orchestration, deployment.
- Ship **Agentic RAG services** with schema-validated outputs, guardrails, and cost-aware routing between models.

Full Stack Developer
01/2023 – 05/2025

MSA Software · Remote

- Cut dashboard response times **1.5–2x** by profiling slow MongoDB aggregation pipelines, adding a caching layer, and lazy-loading heavy panels.
- Moved the team's legacy JavaScript to **TypeScript and Next.js**, removing a whole class of type-related production bugs.
- Migrated shared state from hand-rolled Redux to **Redux Toolkit slices**.

Software Developer (SDE 1)
09/2021 – 01/2023

Manorama Infosolutions · Healthcare software · Remote

- Built REST routing layers for hospital information systems on **ASP.NET + React**; **2x** payload throughput on the hot paths.
- Moved tangled request handling into MVC and hardened it for concurrent user sessions.

SKILLS

AI ENGINEERING

Generative AI · LLMs · RAG · Embeddings & vector search · Prompt engineering · Fine-tuning / distillation · Agentic workflows · Function calling · MCP · LLM evals · OpenAI · Gemini · Claude APIs · PyTorch · Computer vision (YOLOv8) · Ollama · pgvector · Vectorize

FRONTEND

JavaScript · TypeScript · React · Next.js · Redux Toolkit · Zustand · React Flow · React Native / Expo · Tailwind CSS · HTML / CSS

BACKEND & DATA

Node.js · Express · REST APIs · Python · FastAPI · Rust / Tauri 2 (using Claude Code) · Java / Spring Boot · ASP.NET · MongoDB · PostgreSQL · SQL Server · SQLite · Firestore · Redis · WebSockets · JWT auth

DELIVERY

Git · GitHub Actions · CI/CD · Docker · Cloudflare Workers · GitHub Pages · Unit & integration testing · System design · Agile delivery

EDUCATION

B.E. Computer Science · 2021
D. Y. Patil College of Engineering

Diploma, Computer Science · 2018
D. Y. Patil Polytechnic

Certifications — Coding Ninjas: Front End Architecture · React Production Development · Advanced Backend Systems

FitRank — Typed AI decisions for staffing

Applied ML · Live

Ranks people for open tasks with typed decisions instead of a chatbot's opinion. Code applies the hard rules; a 23M-parameter MiniLM-L6 student with five linear heads — distilled from Qwen3.5-9B with a KL loss against the teacher's full distributions — returns a probability for every option; thresholds band each person shortlist / review / hidden, and a manager makes every final call. Managers' accepts and rejects become the next model's training data.

PROOF Evaluated against a code-only baseline on every release (35 held-out tasks, synthetic data): **hit@5 94% vs 91%**, **pairwise 0.88 vs 0.86**, NDCG@10 0.89 vs 0.89 — and **hit@1 54% vs 74%**, which is why it shortlists rather than assigns. FastAPI router → service → repository, Postgres + pgvector, Alembic, JWT + refresh-cookie auth with roles, rate limits, a budget guard, Docker, GitHub Actions CI, **93.5% backend test coverage**.

Knowledge Distillation · MiniLM / PyTorch · FastAPI · Postgres + pgvector · React + TypeScript · Docker · CI

Live ↗ Case study ↗

streamerOS — Local-first streaming cockpit

Desktop · Rust

A Rust + Tauri 2 desktop app for Twitch and YouTube streamers. Chat is read locally, velocity and sentiment are scored in Rust off the UI thread against the channel's own baseline, and scene automation drives OBS over WebSocket v5 — no account, no cloud, pull the network cable and it keeps working. Rust-core-to-webview IPC is batched MessagePack rather than per-event JSON; the assistant runs on Ollama at 127.0.0.1.

PROOF Rust written with Claude Code, reviewed and tested by me. **187 commits** since May 2026, 70 Rust files / **24,292 lines of Rust**, 42,733 lines of TypeScript, **306 Rust tests**. The app repository is private; the marketing site and its Tier-1 support agent — a Hono router on Cloudflare Workers that embeds the product docs into Upstash Vector and streams a gemini-flash answer constrained to that context — are public.

Rust · Tauri 2 · OBS WebSocket v5 · MessagePack IPC · Ollama · Next.js

Website ↗ Case study ↗ Site + agent source ↗

Portfolio Concierge — Grounded RAG API + MCP server

AI Infrastructure · Live

One zero-dependency Cloudflare Worker, two interfaces. A grounded RAG API answers questions about my work from a Vectorize index of the site's 80-document corpus (34 articles, 46 project pages) and refuses below a similarity threshold rather than guess. A public MCP server exposes search_portfolio, get_products and get_availability — any recruiter can add the URL to Claude and interview this résumé from inside their own AI. The corpus re-embeds automatically after every deploy.

Cloudflare Workers · Vectorize · Workers AI · MCP · Grounded RAG · TypeScript

Try it ↗ MCP ↗ Code ↗

IntegrateX — Workflow automation engine

React Flow · Live Sandbox

A node-based workflow canvas — triggers, processing layers, directional edge bindings — backed by a Zustand store. The naive design stored the React Flow render model; a custom Serialization Adapter separates the view from the transport record and cut stored payloads by **94%**, round-trip lossless. The architecture runs live in the browser: drag nodes, rewire the graph, watch the serialization bench.

React Flow · Zustand · TypeScript · Serialization Adapter · MongoDB

Live sandbox ↗ Write-up ↗

Sable — Local-first AI financial agent

Mobile · AI Agent

A React Native / Expo personal-finance agent where the model is structurally incapable of touching the money. 100% of financial data stays on-device in SQLite with no cloud backend; an OpenAI function-calling agent tracks peer-to-peer debts and pushes proactive briefings, and every database mutation renders as a Review & Confirm card — it proposes, only a human commits. Bank-SMS ingestion uses the Adapter pattern with a p-queue serialising SQLite writes.

React Native / Expo · SQLite · OpenAI Function Calling · Local RAG · TypeScript

Case study ↗

APPLIED & CLIENT WORK

ShelfSight AI — Retail vision for a kirana shop

Computer Vision · Open Source

Four retail vision problems in one on-premise system: phantom stock, planogram compliance, produce freshness and expiry-date OCR. A manager lays out a rack once; after that a photo counts every facing and staff are told which shelf to walk to. Its most useful property is what it refuses to claim — the detector localises products without naming them, so the planogram endpoint returns 409 rather than a score it cannot support. **650 tests**.

FastAPI · YOLOv8 · MobileNetV2 · Next.js 14 · Docker · Ollama

Backend ↗ Dashboard ↗

AISA — Student association platform

Full-Stack · Live

Arrived as one 4,019-line HTML file with a live Firebase key in the source and every permission decided in the browser. Rebuilt as a Next.js static export on GitHub Pages and a Spring Boot 3 / Java 25 API on Render, both on Firestore. Firebase Auth answers who you are; the server decides what you may do — the half the original had backwards. Five auth states, **29 tests**.

Next.js · Java 25 · Spring Boot 3 · Firestore · Firebase Auth

Live ↗ Frontend ↗ Backend ↗

Path Saathi LMS — Vocational-skilling platform

Client · 1-Day Delivery

A free vocational-skilling LMS for a client sitting on a library of course videos and PDFs. Brief and MVP sketch arrived on a Monday; a working platform — auth, course delivery, self-paced flow — was live on the dev environment the next day.

Next.js · React · Tailwind · TypeScript

Live (dev) ↗

Autonomous Content Machine — Three pipelines, zero servers

Pipelines · $0 Infra

Three cooperating pipelines on GitHub Actions, running unattended at no cost: an AI writer that drafts a technical article with a brand-locked cover on Mon/Wed/Fri, commits it and deploys the static site; a LinkedIn agent that turns each post into scheduled, UTM-tracked content Tue–Thu with its state ledger committed to Git; and a nightly DEV.to syndicator that cross-posts with canonical URLs pointing home.

GitHub Actions · Python · Next.js · DEV.to API · Zero-Cost Infra

Case study ↗ LinkedIn agent ↗

WRITING & TEACHING

Engineering blog

34 articles

First-person write-ups of the systems above — serialization adapters, grounded RAG, agent tracing with OpenTelemetry, prompt caching, WebSocket telemetry at scale.

Read the blog ↗

DEV.to

35 articles

Syndicated automatically every night with canonical URLs pointing home.

dev.to/yaseenyk04 ↗

AI Engineering Journey

16 days · 64 hours

A curriculum I authored — embeddings, vector search, RAG, agents, evaluation, fine-tuning, production — as 16 interactive chapters with diagrams and a built-in presentation mode.

Open the curriculum ↗

Open source

54 public repos

Since 2022, including the full source of this portfolio, its RAG/MCP worker, the content pipelines, ShelfSight and AISA.

github.com/Yaseenyk ↗